

No Integer Stretch Factor for a Pseudo-Anosov Map

Disproof of Conjecture 000000000088

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Verdict. The conjecture is false at the prime $p = 2$. Every pseudo-Anosov stretch factor is an algebraic unit, whereas the rational integer 2 is not an algebraic unit.

The arithmetic contradiction. An algebraic unit is an algebraic integer whose reciprocal is also an algebraic integer. If 2 were a pseudo-Anosov stretch factor, then $1/2$ would be an algebraic integer. A rational algebraic integer must be an integer: if a reduced rational number a/b is a root of a monic polynomial in $\mathbb{Z}[x]$, the rational-root theorem gives $b = 1$. But $1/2$ is not an integer. Therefore no pseudo-Anosov element has stretch factor 2. Since 2 is prime and the conjecture asserts existence for every prime $p \geq 2$, this one case disproves it.

Published theorem used. We use the classical result attributed to Thurston that each pseudo-Anosov stretch factor is an algebraic unit. See W. P. Thurston, *On the geometry and dynamics of diffeomorphisms of surfaces*, *Bull. Amer. Math. Soc.* 19 (1988), no. 2, 417–431, doi:10.1090/S0273-0979-1988-15685-6. For an explicit published statement of the algebraic-unit consequence and its attribution, see J. Pankau, *Salem number stretch factors and totally real fields arising from Thurston’s construction*, *Geom. Topol.* 24 (2020), 1695–1716, published article.

Lean correspondence and limitation. The Lean 4 project proves the integer-unit lemma and a conditional obstruction theorem. Its explicit parameter ‘unitBridge’ records the precise arithmetic consequence supplied by the external topology and number-theory results: whenever a hypothetical witness has stretch factor 2, the resulting algebraic-unit argument gives an integer reciprocal for 2. Lean proves that such an integer-unit value can only be 1 or -1 , hence cannot be 2.

The Lean project does not formalize mapping class groups, pseudo-Anosov maps, Thurston’s theorem, or the theorem that rational algebraic integers are integers. The original conjecture is disproved by the written proof using those cited published results; Lean verifies the arithmetic deduction after the bridge is supplied as an explicit theorem parameter. Thus the submission does not claim an end-to-end Lean formalization of the original statement. The project introduces no custom axioms, `sorry`, or `native_decide`.

Reproduction. The accompanying self-contained project uses Lean 4.33.1 and core Lean. From the ‘lean’ directory, run ‘lake build’, then ‘lake env lean -DwarningAsError=true Main.lean’. The file prints the axiom dependencies of its main theorems.